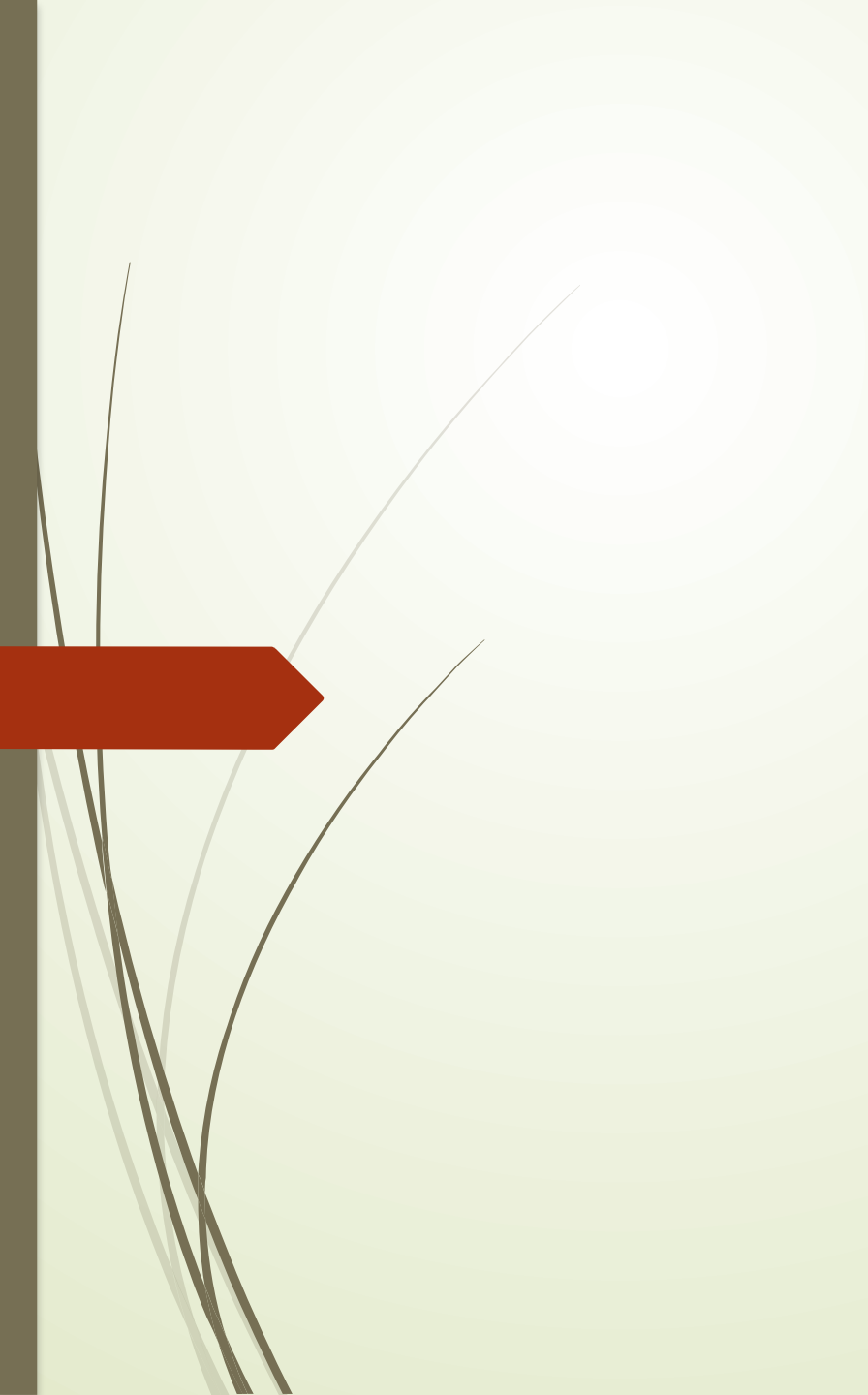
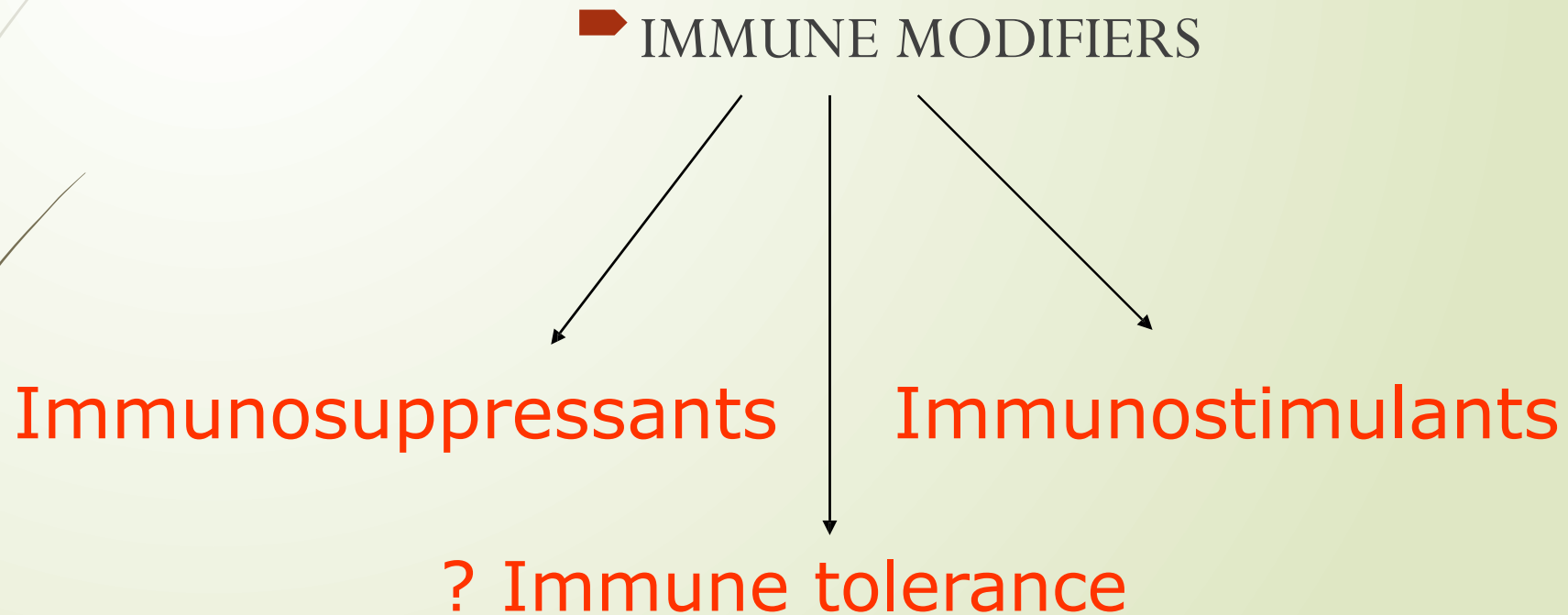




Immunopharmacology

Solomon A.

Immunomodulators





Immunopharmacology

- Agents that suppress the immune
- Agents that augment the immune response or selectively alter the balance of various components of the immune system
- Immune system evolved to differentiate between self from non-self
- Robust receptor mediated sensing and effector mechanism
 - ✓ Innate
 - ✓ Adaptive
- Immune-mediated diseases are significant medical problems



Immuno...

- ☐ Adaptive immune system is mobilized by cues
- ☐ The adaptive immune system has a number of characteristics:
 - ✓ Respond to a variety of antigens
 - ✓ Discriminate between “non-self” and self antigens of the host
 - ✓ Respond to a previously encountered antigen in a learned way by initiating a vigorous memory response
 - ✓ Production of **antibodies**, and the activation of **T lymphocytes**,
 - ✓ Professional **APCs** play pivotal roles



The Immune Response - why and how ?

- Discriminate: Self / Non self
- Destroy:
 - Infectious invaders
 - Dysregulated self (cancers)
- Immunity:
 - Innate, Natural
 - Adaptive, Learned

Who are involved ?

○ Innate

- Complement
- Granulocytes
- Monocytes/macrophages
- NK cells
- Mast cells
- Basophils

○ Adaptive:

- B and T lymphocytes
- B: antibodies
- T : helper, cytolytic, suppressor.

ABNORMAL IMMUNE RESPONSES

- ☐ Hypersensitivity
- ☐ Autoimmunity
- ☐ Immunodeficiency Diseases

IMMUNOSUPPRESSIVE AGENTS

- Dampen immune response in organ transplantation and autoimmune disease
- Particularly important for transplantation, autoimmune disorders, allergies, and all the cases where immune system is too active.
 - ✓ Glucocorticoids
 - ✓ Calcineurin inhibitors
 - ✓ Antimetabolites
 - ✓ Biologicals



Immunosuppressants

- Organ transplantation
- Autoimmune diseases

Problem



- Life long use
- Infection, cancers
- Nephrotoxicity
- Diabetogenic



Principle

- Transplantation needs
 - ✓ Careful patient selection
 - ✓ ABO blood typing
 - ✓ Multitiered immunotherapy
 - ✓ Intensive induction and low dose maintenance
 - ✓ Evaluation for rejection
 - ✓ Drug withdrawal or de-escalation if any toxicity

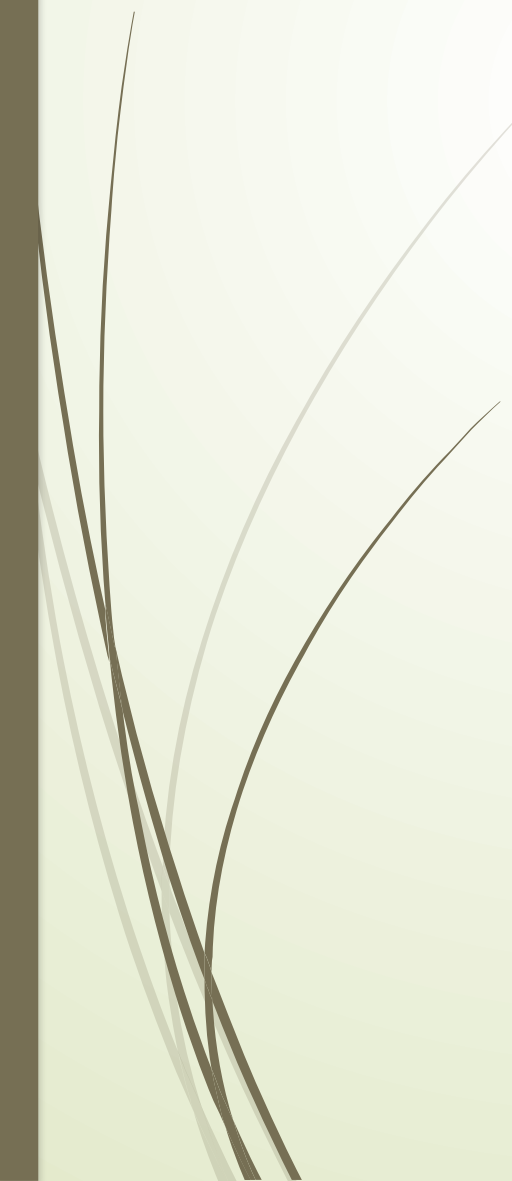
Source	Immunopharmacologic Agents Used	Response
Autoimmune diseases		
Idiopathic thrombocytopenic purpura (ITP)	Prednisone, ¹ vincristine, occasionally cyclophosphamide, mercaptopurine, or azathioprine; commonly high-dose gamma globulin, plasma immunoadsorption or plasma exchange	Usually good
Autoimmune hemolytic anemia	Prednisone, ¹ cyclophosphamide, chlorambucil, mercaptopurine, azathioprine, high-dose gamma globulin	Usually good
Acute glomerulonephritis	Prednisone, ¹ mercaptopurine, cyclophosphamide	Usually good
Acquired factor XIII antibodies	Cyclophosphamide plus factor XIII	Usually good
Autoreactive tissue disorders (autoimmune diseases) ²	Prednisone, cyclophosphamide, methotrexate, interferon- α and - β , azathioprine, cyclosporine, infliximab, etanercept, adalimumab	Often good, variable
Isoimmune disease		
Hemolytic disease of the newborn	Rh ₀ (D) immune globulin	Excellent
Organ transplantation		
Renal	Cyclosporine, azathioprine, prednisone, ALG, OKT3, tacrolimus, basiliximab, ³ daclizumab, ³ sirolimus	Very good
Heart	Cyclosporine, azathioprine, prednisone, ALG, OKT3, tacrolimus, basiliximab, ³ daclizumab, ³ sirolimus	Good
Liver	Cyclosporine, prednisone, azathioprine, tacrolimus, sirolimus	Fair
Bone marrow	Cyclosporine, cyclophosphamide, prednisone, methotrexate, ALG	Good
Prevention of cell proliferation		
Coronary stents	Sirolimus (impregnated stent)	Good
Neovascular macular degeneration	Ranibizumab (labeled), bevacizumab (off-label)	Fair

GLUCOCORTICOIDS

- Administration of any glucocorticoid reduces the size and lymphoid content of the lymph nodes and spleen
- Cellular immunity is more affected
- Increases the fractional catabolic rate of IgG, GC receptors or induced proteins regulate transcription
- Induce apoptosis
- Pro-inflammatory cytokines are downregulated



Gluc...

- High dose of methylprednisolone for acute organ rejection
 - For autoimmune conditions such as MS
- 

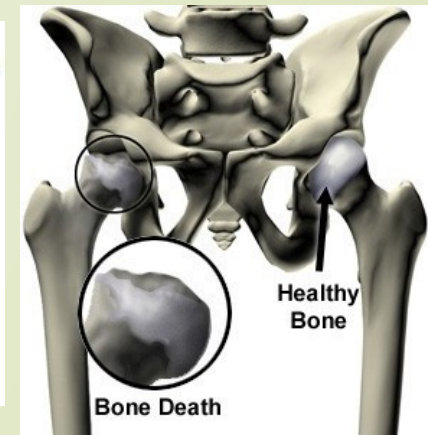
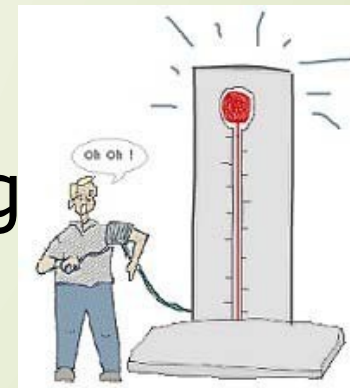
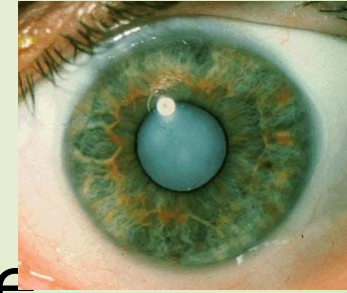


USES - Glucocorticoids

- Transplant rejection
- GVH – BM transplantation
- Autoimmune diseases – RA, SLE, Hematological conditions
- Psoriasis
- Inflammatory Bowel Disease, Eye conditions

Toxicity

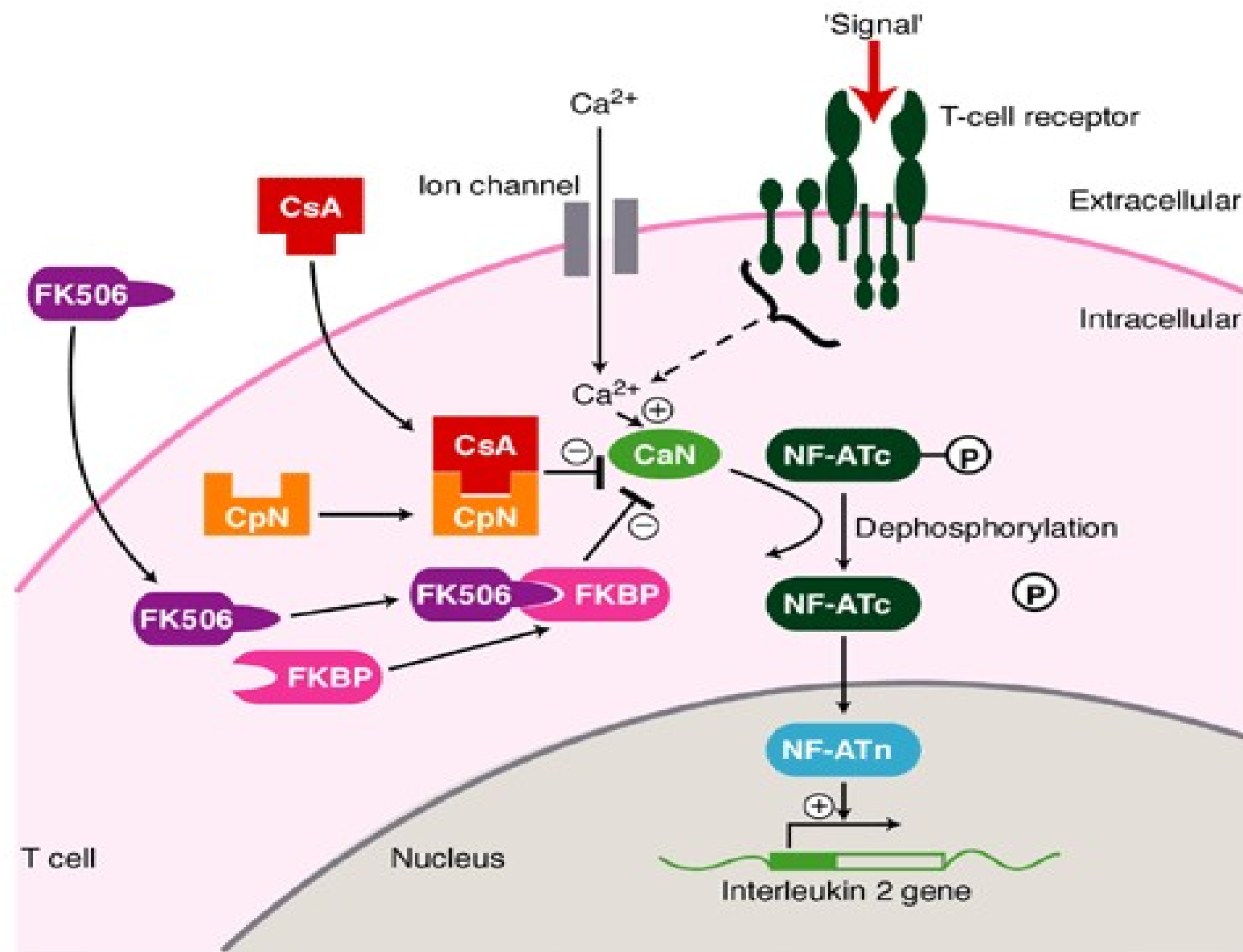
- Growth retardation
- Avascular Necrosis of Bone
- Risk of Infection
- Poor wound healing
- Cataract
- Hyperglycemia
- Hypertension





Calcineurin inhibitors

- Cyclosporine
 - Tacrolimus
 - Most effective immunosuppressive drugs
 - Target intracellular signaling pathways
 - Blocks Induction of cytokine genes
- 



Mechanism of action of cyclosporine or tacrolimus (FK506)



Cyclosporine

- More effective against T-cell dependent immune mechanisms — transplant rejection, autoimmunity
- IV, Oral

Uses

- Organ transplantation: Kidney, Liver, Heart
- Rheumatoid arthritis, IBD, uveitis
- Psoriasis
- Aplastic anemia
- Skin Conditions- Atopic dermatitis, Alopecia Areata

Toxicity : Cyclosporine

- Renal dysfunction
- Tremor
- Hirsutism
- Hypertension
- Hyperlipidemia
- Gum hyperplasia
- Hyperuricemia – worsens gout
- Calcineurin inhibitors + Glucocorticoids = Diabetogenic



Drug Interaction : Cyclosporine

○ CYP 3A4

- Inhibitors: CCB, Antifungals, Antibiotics, HIV PI, Grape juice
- Inducers: Rifampicin, Phenytoin
- Additive nephrotoxicity: NSAIDs



Tacrolimus

- Inhibits T-cell activation by inhibiting calcineurin
- Immunosuppressant macrolide antibiotic
- Use
 - Prophylaxis of solid-organ allograft rejection



Toxicity - Tacrolimus

- Nephrotoxicity
- Neurotoxicity-Tremor, headache, motor disturbances, seizures
- GI Complaints
- Hypertension
- Hyperglycemia
- Risk of tumors, infections
- Drug interaction
 - Synergistic nephrotoxicity with cyclosporine
 - CYP3A4



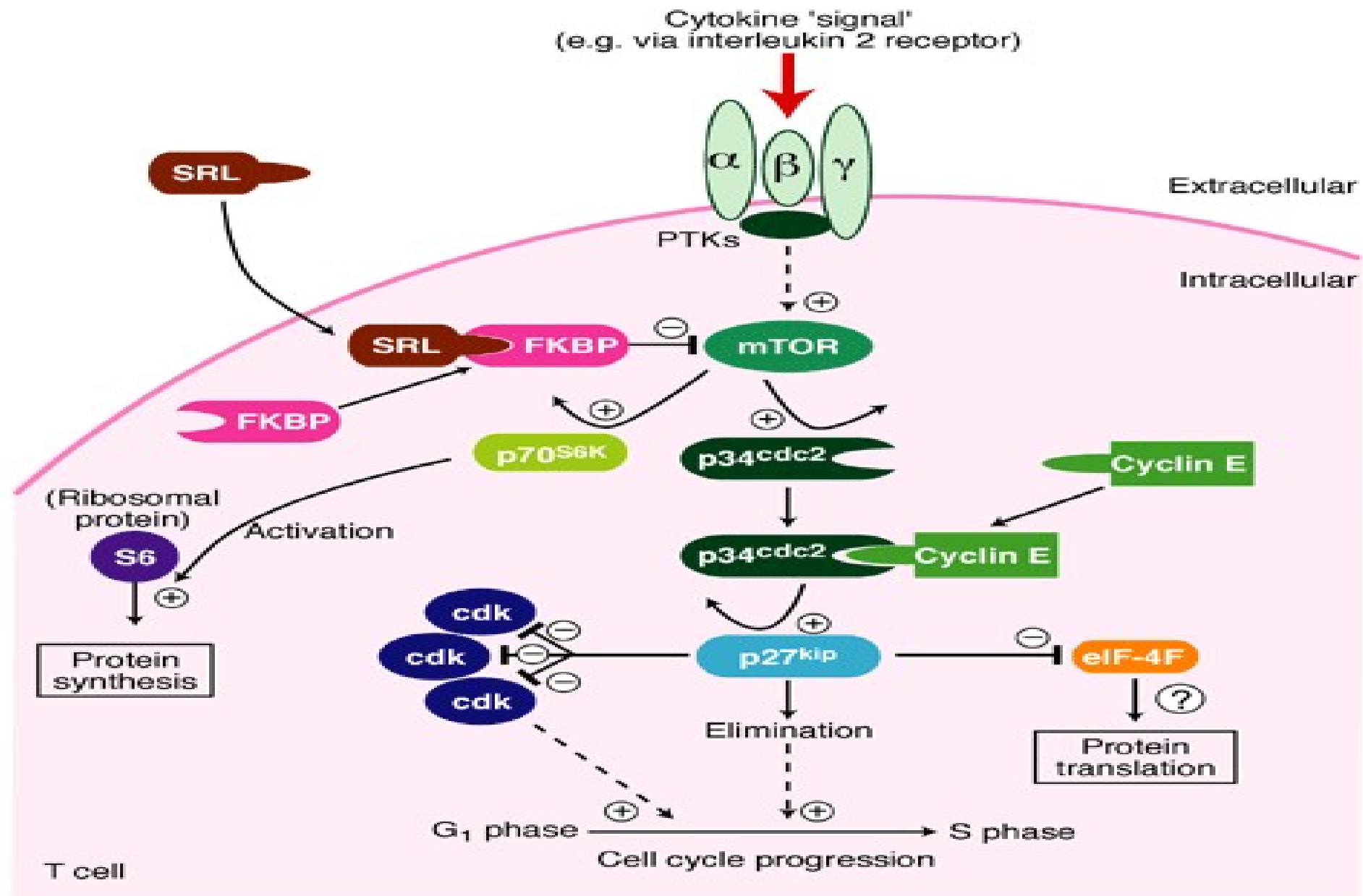
Antiproliferative and Antimetabolic drugs

- Sirolimus
- Everolimus
- Azathioprine
- Mycophenolate Mofetil
- Others:
 - Methotrexate
 - Cyclophosphamide
 - Thalidomide
 - Chlorambucil
 - Leflunomide
 - Lenalidomide



Sirolimus

- Inhibits T-cell activation and Proliferation
- Complexes with an immunophilin, Inhibits a key enzyme in cell cycle progression – mammalian target of rapamycin (mTOR)



Mechanism of action of sirolimus (rapamycin)



Sirolimus

Uses

- Prophylaxis of organ transplant rejection along with other drugs

Toxicity

- Increase in serum cholesterol, Triglycerides
- Anemia
- Thrombocytopenia
- Hypokalemia
- Fever
- GI effects
- Risk of infection, tumors

MYCOPHENOLATE MOFETIL

- Semisynthetic derivative of mycophenolic acid
- Inhibits de novo synthesis of purines
- a selective non-competitive reversible inhibitor of IMPDH
- MPA inhibits lymphocytes proliferation and function
- Used in combination for prophylaxis of graft rejection



Cytotoxics

- Azathioprine
- Prodrug of mercaptopurine
- Interfering with purine nucleic acid metabolism
- For acute glomerulonephritis and in the renal component of systemic lupus erythematosus.
- Rheumatoid arthritis, Crohn's disease, and multiple sclerosis

Hydroxychloroquine

- ❑ Has immunosuppressant properties.
 - Suppress intracellular antigen processing and loading of peptides onto MHC class II molecules
 - Decrease T-cell activation.
 - Used to treat some autoimmune disorders, eg, rheumatoid arthritis and systemic lupus erythematosus

Immunosuppressant antibodies

Muromonab CD3

- ✓ Murine monoclonal antibody against the CD3 glycoprotein
- ✓ Obstructs the binding of MHC II-antigen complex to the T cell receptor:
- ✓ Binding to the CD3 protein results in a disruption of T-lymphocyte function
- ✓ less broad immunosuppression
- Muromonab CD3 has been used as induction therapy together with corticosteroids and azathioprine with delayed use of cyclosporine in 'sequential regimen' for organ transplantation. This serves to postpone potential nephro- and hepatotoxicity of cyclosporine.



Adalimumab

- Human IgG1
- A monoclonal antibody that binds specifically to TNF alpha, blocking its interaction with cell surface TNF receptors.
- Rheumatoid arthritis, juvenile idiopathic arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, and plaque psoriasis.
- Adalimumab lyses cells expressing TNF
- Reduces levels of C-reactive protein, erythrocyte sedimentation rate, serum IL-6, and matrix metalloproteinases MMP-1 and MMP-3



Certolizumab pegol

- Recombinant humanized Fab fragment that binds to TNF- α .
- It is coupled to a 40-kDa polyethylene glycol.
- ✓ Neutralizes the activity of membrane-associated and soluble TNF- α without lysing cells.
- ✓ Certolizumab is indicated for patients with Crohn's disease and rheumatoid arthritis.

Etanercept

- ✓ Dimeric fusion protein ... IgG1 constant regions and TNF receptor
- ✓ Binds to both TNF- α and TNF- β
- ✓ Similar to those of adalimumab and infliximab, ie, inhibition of TNF- α -mediated inflammation, but its half-life is shorter due to its physical form and the route of injection
- ✓ Etanercept is approved for adult rheumatoid arthritis, polyarticular juvenile idiopathic arthritis, ankylosing spondylitis and psoriatic arthritis.
- ✓ It may be used in combination with methotrexate in some patients with arthritis



Immunostimulants

- Levamisole
- Thalidomide
- BCG
- Recombinant Cytokines
 - Interferons
 - Interleukin-2



Levamisole

- Antihelminthic
- Restores depressed immune function of B, T cells, Monocytes, Macrophages
- Adjuvant therapy with 5FU in colon cancer

Toxicity

- Agranulocytosis



Thalidomide

- Birth defect
- Contraindicated in women with childbearing potential
- Enhanced T-cell production of cytokines – IL-2, IFN- γ
- NK cell-mediated cytotoxicity against tumor cells

USE:

- Multiple myeloma

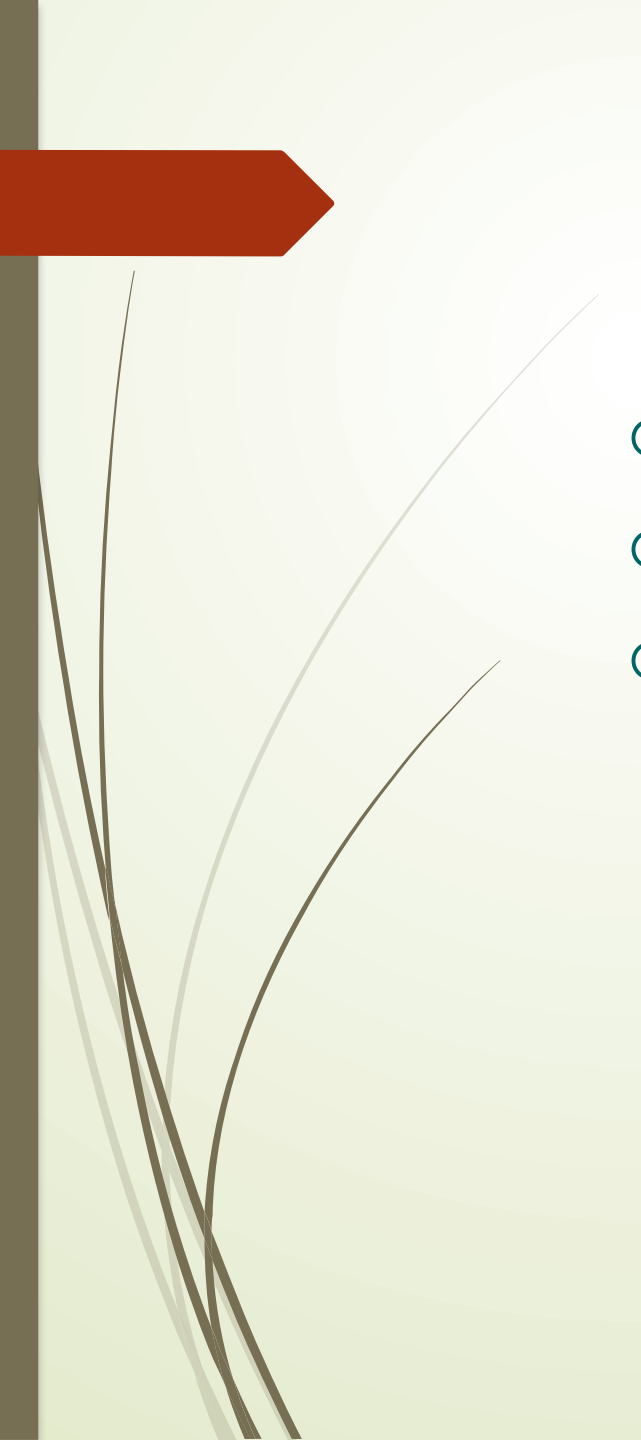


Bacillus Calmette-Guerin

- Live, attenuated culture of BCG strain of Mycobacterium Bovis
- Carcinoma Bladder

Adverse Effects

- Hypersensitivity
- Shock
- Chills



Interferons

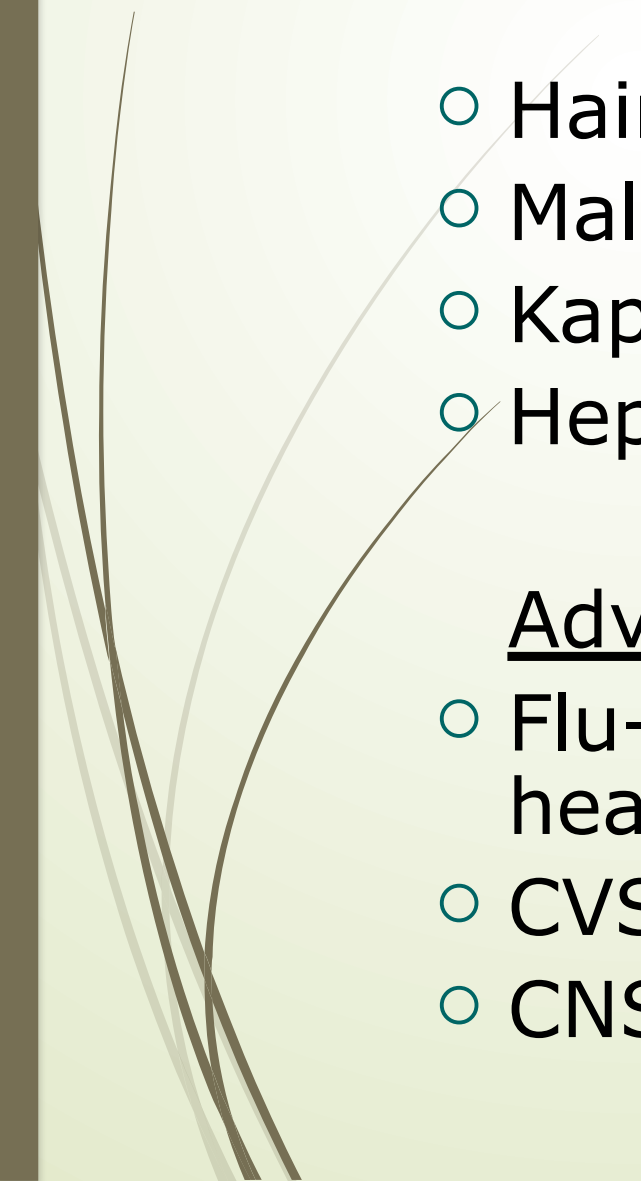
- Antiviral
- Immunomodulatory activity
- Bind to cell surface receptors – initiate intracellular events
 - Enzyme induction
 - Inhibition of cell proliferation
 - Enhancement of immune activities
 - Increased Phagocytosis



Interferon alpha-2b

- Hairy cell leukemia
- Malignant melanoma
- Kaposi sarcoma
- Hepatitis B

Adverse reactions

- Flu-like symptoms – fever, chills, headache
 - CVS- hypotension, Arrhythmia
 - CNS- depression, confusion
- 

Interleukin-2 (aldesleukin)

- Proliferation of cellular immunity – Lymphocytosis, eosinophilia, release of multiple cytokines – TNF, IL-1, IFN- γ
- Uses
- Metastatic renal cell carcinoma
- Melanoma
- Toxicity
- Cardiovascular: capillary leak syndrome, Hypotension

Filgrastim

- Recombinant human G-CSF
- It is a nonglycosylated peptide of 175 amino acids,
- Filgrastim is a short-acting recombinant G-CSF that mimics the biological actions of endogenous G-CSF.
- It also facilitates the release of neutrophils from the bone marrow into the blood to reduce the incidence of infection and manage neutropenia.